黄璐(Huang Lu)

香港

电话：13870728762 | 邮箱：hl13870728762 @outlook.com

GitHub: github.com/lulu5132

个人总结

- AI应用开发方向的数据科学硕士，具备物联网工程与人工智能交叉背景。专注于 **大语言模型（LLM）应用开发与系统工程落地**，熟悉 **RAG 架构、Prompt Engineering 与 AI Agent 编排**，能够从 0 到 1 搭建实际场景的 AI 工具。
- 具备扎实的 **Python、机器学习与强化学习基础**，曾获多项国家级 AI 竞赛奖项，并拥有**实用新型专利（第一发明人）**。在自动驾驶、法律合规分析等复杂场景中具备 **AI系统设计与工程实现经验**。

教育背景

岭南大学 (Lingnan University) **数据科学硕士 (人工智能学部)** **2025.09 - 2026.06**

- 人工智能学科全球 **151-200（软科世界一流学科排名）**

上海商学院 **物联网工程本科** **2021.09 - 2025.06**

- **荣誉：上海市优秀毕业生 (Top 5%)**、连续两年校**"优秀共青团员"**、三次校综合二等奖学金。

核心项目经历

《基于 DeepSeek 的法律合规分析与专利撰写 AI 助手》 **项目负责人** **2026.03 - 至今**

构建面向 **法务与知识产权场景的大模型应用系统**，实现专利提案初稿生成与法律合规分析的自动化。

核心贡献：

- **RAG架构设计：**基于 LlamaIndex + FAISS 构建私有向量数据库，通过 Hybrid Search 提升法律文本检索准确率并降低 LLM 幻觉问题
- **模型协同架构：**实现 DeepSeek-V3/R1 云端 API 与 Ollama 本地模型的动态切换，兼顾推理性能与数据隐私
- **结构化输出控制：**通过 JSON Schema 约束模型输出，实现专利草案到标准化提案文档的自动化转换
- **性能评估：**对不同量化模型在法律术语理解与推理延迟方面进行实验评估，为私有化部署提供选型参考

- **技术栈：**Python, Llama-Index, DeepSeek, FAISS, JSON Schema, Prompt Engineering.
- **GitHub:** https://github.com/lulu5132/legal-rag-assistant

《Hybrid-Traj-Guard：自动驾驶异步协同决策架构设计》 **项目负责人** **2025.12 - 至今**

核心贡献：

- 设计 **LLM（决策层）+ CNN/RNN（执行层）**的异步协同架构，解决 LLM 推理延迟问题
- 复现ICLR论文的基础上，融合 **PPO 强化学习算法与 Barrier Function**，在轨迹规划阶段实现安全约束修正
- 实现 **轨迹预测 + 语义意图解释** 双输出模式，提高复杂决策过程的可解释性
- 相较 baseline 方法 **碰撞率降低约 40%–60%**，平均轨迹误差 ≤ 0.2m

《高并发电商数据库一致性测试》 **测试负责人** **2025.10 - 2025.11**

研究高并发电商场景下数据库事务一致性问题。

核心贡献：

- 编写 Python 自动化脚本模拟 **500+ 并发用户秒杀请求**
- 分析系统 **QPS、响应时间与错误率** 等性能指标
- 验证数据库事务隔离级别，确保高并发情况下 **零超卖**

实习经历

一起美（上海）网络科技有限公司 **用户运营实习生** **2025.06 - 2025.08**

- 构建用户转化数据追踪模型，分析用户行为路径
- 优化小红书 / 抖音标签策略，使视频播放量从 **1.2万提升至 4.2万（+253%）**

荣誉与成果

竞赛奖项：

- 中国机器人及人工智能大赛 国赛一等奖、三等奖
- 全国人工智能应用创新大赛 国赛三等奖
- 中国国际大学生创新大赛 上海赛区铜奖

学术成果：

- 实用新型专利：一种基于物联网的数据传输显示设备（第一发明人）
- 软件著作权：后端数据安全防护系统、前端数据备份存储系统

技术能力

- AI/LLM: RAG 架构，Prompt Engineering，AI Agent 开发，Llama-Index，Transformer / PyTorch
- 算法与机器学习: 强化学习（PPO），K-Means 聚类，轨迹预测与路径规划
- 工程与开发: Cloud GPU Computing，Python（Pandas / NumPy），SQL，Git，FAISS

LU HUANG

Location: Hong Kong

Phone: (+86) 13870728762 | **Email:** hl13870728762 @outlook.com

GitHub: github.com/lulu5132

PROFESSIONAL SUMMARY

- **AI Application Engineer** and M.Sc. candidate in Data Science with an interdisciplinary background in IoT and AI.
- Specializing in **Large Language Model (LLM)** application development, including **RAG architectures, Prompt Engineering**, and **AI Agent orchestration**
- Proven track record in building production-ready AI tools from 0 to 1, with deep expertise in Python, Machine Learning, and Reinforcement Learning.
- Recipient of multiple **National AI Competition Awards** and holder of a **Utility Model Patent** (1st Inventor).

EDUCATION

Lingnan University **M.Sc. in Data Science(School of AI)** **Sep 2025 - Jun. 2026(Expected)**

- (Ranked **151-200 globally** in Artificial Intelligence by ShanghaiRanking's GRAS 2025)

Shanghai Business School **B.Eng. in IoT Engineering** **Sep 2021 -Jun. 2025**

- **Honors:** Shanghai Outstanding Graduate **(Top 5%)**, 3×University Comprehensive Second-Class Scholarships.

KEY PROJECT EXPERIENCE

DeepSeek-based Legal Compliance & Patent Drafting AI Assistant **Project Leader** **Mar 2026 - Present**

- Developed an intelligent system for legal and IP departments to automate patent drafting and compliance analysis.
- **RAG Architecture:** Built a private vector database using **LlamaIndex + FAISS;** implemented **Hybrid Search** to enhance retrieval precision and significantly mitigate LLM hallucinations.
- **Model Orchestration:** Designed a flexible switching mechanism between **DeepSeek-V3/R1 Cloud APIs** and **Ollama local models** to balance performance with data privacy.
- **Structured Control:** Utilized **JSON Schema** to enforce output constraints, enabling seamless transformation of raw drafts into standardized patent documents.
- **Tech Stack:** Python, Llama-Index, DeepSeek, FAISS, JSON Schema, Prompt Engineering.
- **GitHub:** https://github.com/lulu5132/legal-rag-assistant

Hybrid-Traj-Guard: Asynchronous Collaborative Architecture for Autonomous Driving **Project Leader** **Dec 2025 - Present**

- Designed an asynchronous framework combining **LLM (Decision Layer)** with **CNN/RNN (Execution Layer)** to eliminate LLM inference latency in real-time environments.
- Integrated **PPO Reinforcement Learning** and **Barrier Functions** into the trajectory planning phase to enforce physical-level safety constraints.
- Achieved a dual-output mode **(Trajectory + Semantic Intent)** to enhance the explainability of complex AI decision-making.
- **Results:** Reduced collision rates by **40%–60%** compared to baselines; achieved Average Displacement Error **(ADE) ≤ 0.2m.**

High-Concurrency E-commerce Database Consistency Testing **Testing Lead** **Oct 2025 - Nov 2025**

- Conducted stress tests to simulate 500+ concurrent users in "Flash Sale" scenarios using Python automation scripts.
- Evaluated system performance through metrics including **QPS, Response Time (RT)**, and error rates.
- Validated database transaction isolation levels to ensure **"Zero Overselling"** under peak load conditions.

INTERNSHIP EXPERIENCE

Yi Qi Mei (Shanghai) Network Technology Co., Ltd. **User Operations Intern** **Jun 2025 - Aug 2025**

- Built data tracking models to analyze user conversion paths and behavior.
- Optimized social media tagging strategies (Xiaohongshu/Douyin), increasing video views from 12k to 42k **(+253%).**

HONORS & SKILLS

- **Competitions:**
  - **National First & Third Prizes**, The 27th China Robotics & AI Competition
  - **National Third Prize**, The 1st China AI Application Innovation Competition
  - Bronze Medal, China International "Internet+" Undergraduate Innovation & Entrepreneurship Competition
- **Academic Output:**
  - **Utility Model Patent:** An IoT-based Data Transmission Display Device (**1st Inventor**)
  - **Software Copyrights:** Backend Data Security Shield, Frontend Data Backup & Storage System
- **Certifications: MeterSphere Certified Software Testing Engineer**, IoT Technology Level II, Computer Rank Examination Level II (WPS Office).
- **Technical Stack:**
  - **AI/LLM:** RAG, Prompt Engineering, Agent Development, Llama-Index, PyTorch, Transformers.
  - **Algorithms:** Reinforcement Learning (PPO), K-Means Clustering, Path Planning.
  - **Engineering:** Cloud GPU Computing (RTX 4090), Python (Pandas/NumPy), SQL, Git, FAISS.
  - **Languages:** IELTS 6.0 (English-Medium Instruction), Mandarin (Advanced)